Anatomy of a Signal: Dissecting Non-Linear Factor Efficacy in US Equities

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Setup of the Project & Motivation

Modern equity research operates in what is often called the *Factor Zoo*. Researchers now work with hundreds or even thousands of candidate signals. These signals come from fundamentals, technical indicators, and high, frequency trading data. While this has expanded the search for alpha, it has also created a serious problem. Many strategies that look profitable may simply be taking hidden market risk. The core motivation of this project is simple: can modern non, linear machine learning actually identify true idiosyncratic alpha, or does it mostly repackage beta?

To study this question, we construct a large, high, dimensional panel dataset using Bloomberg and WRDS. The equity universe consists of 494 static S&P 500 stocks, observed at a daily frequency from January 2005 to December 2014. The universe is kept static on purpose. This avoids complications from index rebalancing and keeps the focus on signal quality rather than universe changes. For each stock and day, we observe 327 features, all normalized using daily cross, sectional z, scores.

The feature set is intentionally broad. It includes microstructure variables derived from tick, level data, such as spreads, liquidity, and order, flow measures. It also includes fundamental variables, based on accounting ratios and balance, sheet information. Finally, a range of technical indicators are included, covering momentum, volatility, volume, and relative strength. In total, the dataset contains roughly 1.1 million daily observations, making it well suited for machine learning methods but challenging for traditional linear models.

The predictive task is framed as a cross, sectional return prediction problem. The model attempts to forecast next, day returns, r_{t+1}, using only information available at time t. We use LightGBM, a gradient boosting machine that can handle non, linear relationships and a large number of predictors. Model training follows a rolling window approach, so that predictions are always made out, of, sample. This helps reduce look, ahead bias, though no setup is ever perfect.

Predicted returns are converted into portfolios using a simple ranking rule. On each day, stocks are ranked by their predicted signal. The strategy goes long the top 10% of stocks and short the bottom 10%. Portfolio weights are chosen to be dollar neutral, so that

$$\sum w_{\text{long}} = \sum w_{\text{short}}.$$

In addition, the portfolio is constrained to be beta neutral, meaning

$$\sum_i w_i \, \beta_i = 0.$$

This constraint is critical. Without it, a strategy may earn returns simply by loading on market risk. Finally, transaction costs of 4 basis points (0.04%) per unit of turnover are applied to reflect realistic trading frictions.

The overall setup is designed to be intentionally strict. If alpha exists, it should survive these controls. If not, the framework should reveal that clearly.

Let's look at the Data and how we cleaned it!

The primary objective of the data manipulation step is to construct a balanced and stable panel suitable for machine learning. High, dimensional models are sensitive to missing data, broken time series, and structural inconsistencies. If these issues are not handled carefully, model behavior becomes unstable and results are difficult to interpret.

To reduce these risks, the final universe is restricted to the top 100 stocks by market capitalization with complete and continuous trading data over the full sample period from January 2005 to December 2014. This selection ensures that each stock has a long, uninterrupted time series. The main trade, off of this decision is the introduction of survivorship bias. This is accepted intentionally. The goal of the project is not to estimate real, world performance, but to isolate the efficacy of the signals and the model under clean conditions. A smaller but well, behaved panel is preferred over a larger, unstable one.

The raw data sources differ significantly in structure and formatting. Bloomberg data arrives primarily in a wide format, with one row per date and many columns per ticker. In contrast, WRDS data, especially fundamentals and TAQ variables, is provided in long or panel format, with multiple observations per ticker and time. A key preprocessing step is therefore temporal alignment across sources. All datasets are aligned at the daily frequency and merged using a consistent date index.

For fundamental data, temporal alignment is handled using "as, of" merging. Accounting variables are matched to stock, days using the most recent information that was publicly available at that time. This prevents future information from leaking into the model. No forward, looking merges are allowed. While this reduces the effective availability of some features, it preserves econometric correctness.

Before modeling, the dataset undergoes extensive structural cleaning. First, "dead columns" are removed. These include variables that are almost entirely missing, constant over time, or populated only for a very short subsample. Such columns add noise, slow

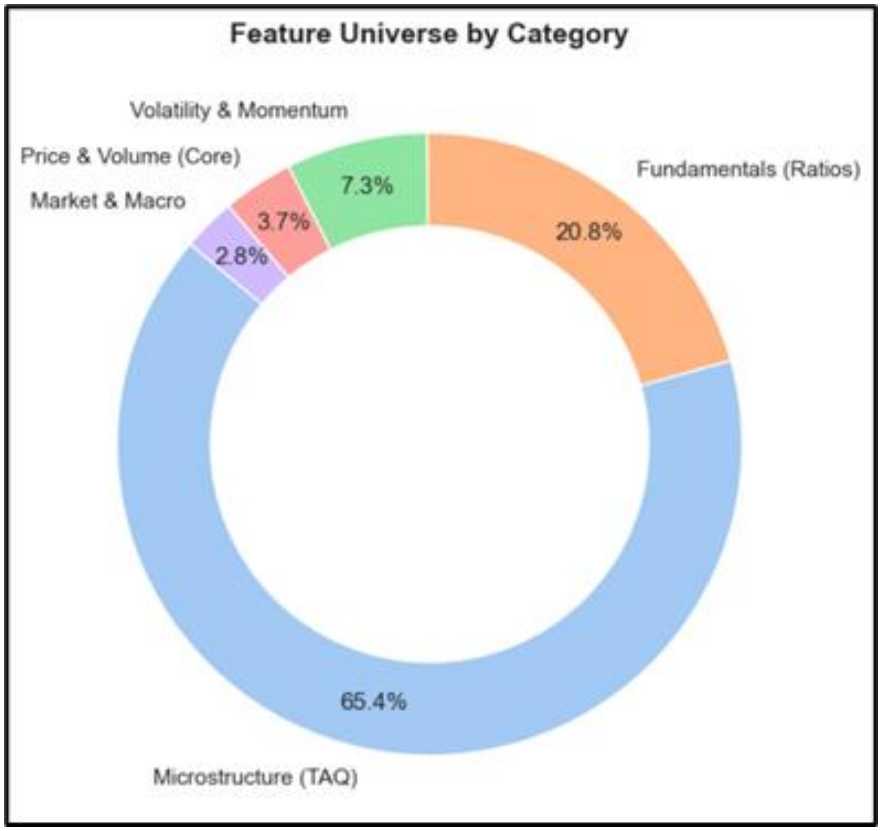
training, and provide no meaningful signal. Next, all variables are explicitly type cast. Many raw inputs are read as strings or mixed types, especially in high, frequency TAQ data. These are carefully converted to numeric formats, with invalid entries set to missing rather than forced.

Missing data is handled conservatively. There is no global imputation using cross, sectional means or medians. This is a deliberate choice. Global filling can introduce artificial structure and leak information across stocks. Limited forward, filling is allowed only within the same stock, and only for short gaps. No variable is overfilled. If information is genuinely unavailable, it remains missing. This approach reflects how real trading data behaves. Importantly, the modeling framework is designed to accommodate this choice. LightGBM naturally handles missing values, learning optimal split directions without requiring aggressive imputation. This allows the preprocessing pipeline to remain simple and robust, without injecting synthetic data that could distort the learning process.

Finally, all features are standardized using daily cross, sectional z, scores. This ensures that variables measured on different scales, such as spreads, ratios, and volumes, are directly comparable. It also prevents the model from implicitly overweighting variables solely due to their units. The data manipulation process prioritizes temporal correctness, structural stability, and transparency. The objective is not to make the data look clean, but to make it honest. Any success or failure of the model should come from the signal itself, not from hidden preprocessing shortcuts.

This is how the feature universe looks like:

Category	**Count**	**Examples**
Microstructure (TAQ)	214	Bid, ask spreads, BidLow...
Fundamentals (Ratios)	68	pe_, debt, cash...
Volatility & Momentum	24	Vol_, Momentum_, IVol...
Price & Volume (Core)	12	Close, Open, High...
Market & Macro	9	SPX, VIX, USGG3M...



Methodology: A Deep Dive

Overview of the Modeling Framework

The core objective of the methodology is to map a high, dimensional vector of firm, level predictors into a scalar score that ranks stocks by their expected future returns. Formally, for each stock $i$on day t, we observe a feature vector

$$\mathbf{x}_{i,t} \in \mathbb{R}^{327},$$

and seek to learn a function

$$f: \mathbb{R}^{327} \to \mathbb{R},$$

such that

$$\hat{y}_{i,t} = f(\mathbf{x}_{i,t})$$

is informative about the stock's forward excess return $r_{i,t+1}$.

Rather than focusing on point prediction accuracy, the problem is framed as a cross, sectional ranking task. The relative ordering of $\hat{y}_{i,t}$across stocks on a given day is the economically relevant output. This framing naturally motivates the use of non, linear machine learning methods that can model interactions across heterogeneous predictors without strong parametric assumptions.

The Model: Gradient Boosted Decision Trees

Decision Trees as Base Learners

A decision tree partitions the feature space into disjoint regions using a sequence of binary splits. Each internal node corresponds to a decision rule of the form

$$x_j \leq c,$$

where $x_j$is a feature and $c$is a threshold. Terminal nodes (leaves) assign a constant prediction value.

A single regression tree approximates the target as

$$f(\mathrm{x}) = \sum_{k=1}^{K} \gamma_k \, \mathbb{I}(\mathrm{x} \in R_k),$$

where $\{R_k\}$are the leaf regions and $\gamma_k$are leaf outputs. While trees are interpretable and flexible, a single tree is typically unstable and has high variance.

Boosting: Sequential Error Correction

Gradient Boosting addresses this limitation by constructing an **additive ensemble** of trees. The model takes the form

$$f_M(\mathbf{x}) = \sum_{m=1}^{M} \nu \, h_m(\mathbf{x}),$$

where $h_m$is the m, th tree and $\nu \in (0,1]$is a learning rate.

Unlike Random Forests, which train trees independently, boosting trains trees **sequentially**. Each new tree is explicitly trained to correct the errors made by the current ensemble. This is achieved by fitting trees to the **negative gradient of the loss function**, evaluated at the current predictions.

Let $L(y, f(\mathbf{x}))$denote the loss function. At iteration m, the pseudo, residuals are

$$r_i^{(m)} = -\frac{\partial L(y_i, f(\mathbf{x}_i))}{\partial f(\mathbf{x}_i)} |_{f=f_{m-1}}.$$

The next tree $h_m$is fit to these residuals. This process can be interpreted as a form of functional gradient descent in function space.

Loss Function and Regression Objective

In this project, the model is trained under a regression objective, where the loss is typically the squared error:

$$L(y, \hat{y}) = (y - \hat{y})^2.$$

Although the economic task is ranking, regression loss is sufficient because the **monotonic ordering** of predictions drives portfolio construction. The continuous output provides more information than a binary or categorical label, which is critical in noisy financial settings.

Why LightGBM?

Computational Efficiency

LightGBM is an optimized implementation of gradient boosting designed for **large, dense datasets**. It introduces several algorithmic innovations that are particularly relevant in this setting:

- **Histogram, based splitting**: Continuous features are discretized into bins, dramatically reducing computational cost.
- **Leaf, wise tree growth**: Trees are grown by splitting the leaf with the largest loss reduction, rather than level, wise growth. This allows LightGBM to capture complex structures with fewer splits.
- **Memory efficiency**: Sparse awareness and optimized data storage allow the model to scale to millions of observations and hundreds of features.

Given the dataset size (~1.1 million observations × 327 features), these properties are essential for feasible training.

Automatic Interaction Detection

A major advantage of tree, based models is their ability to **learn interactions endogenously**. LightGBM can represent rules of the form:

$$\text{If (Volume is High)} \wedge \text{(Volatility is Low)} \Rightarrow \text{Positive Signal.}$$

Such conditional logic is extremely difficult to capture using linear regression without manually specifying interaction terms. In high, dimensional factor spaces, enumerating all possible interactions is infeasible. Gradient boosting implicitly searches this space during training.

Handling Missing Data

LightGBM natively handles missing values by learning optimal default split directions. For a split on feature x_j, missing observations are assigned to the child node that minimizes loss. This eliminates the need for aggressive imputation and allows the model to exploit patterns in missingness itself, which may be economically meaningful.

Training Protocol: Walk, Forward Validation

Look, Ahead Bias in Financial Data

Financial time series exhibit strong temporal dependence. Any training procedure that allows future information to influence past predictions results in look, ahead bias and invalid inference. This risk is especially severe in machine learning models, which can exploit subtle data leakage.

Expanding Window Framework

To prevent this, model training follows an **expanding window (walk, forward) protocol**. Let T_0denote the initial training endpoint. The model is trained on

$$\{(\mathbf{x}_{i,t}, r_{i,t+1}) : t \leq T_0\},$$

and then used to generate predictions for $t = T_0 + 1$. The training window is then expanded, and the process repeats.

This approach mirrors real, time deployment: at each point in time, the model only has access to past information. It also allows the model to adapt gradually as more data becomes available.

Scoring and Cross, Sectional Ranking

Model Output

For each stock $i$on day t, the trained model produces a **raw prediction score**

$$\hat{y}_{i,t}.$$

This score is a continuous scalar that can be interpreted as an estimate of expected excess return. Importantly, its absolute magnitude is not directly meaningful. Only relative ordering matters.

Daily Ranking Mechanism

On each day in the test period (2013–2014), the 100 stocks in the universe are ranked by $\hat{y}_{i,t}$. Formally, define the rank operator:

$$\text{Rank}_t(i) = \text{rank of } \hat{y}_{i,t} \text{ among } i = 1, \dots, 100.$$

- Rank 1 corresponds to the highest conviction long.
- Rank 100 corresponds to the highest conviction short.

This cross, sectional ranking transforms noisy point predictions into a robust relative signal, which is more stable in the presence of low signal, to, noise ratios.

Evaluation Metric: Information Coefficient (IC)

Definition

Model performance is evaluated using the **Information Coefficient (IC)**, defined as the Spearman rank correlation between predicted scores and realized returns:

$$\text{IC}_t = \rho_s\left(\{\hat{y}_{i,t}\}_{i=1}^{N}, \{r_{i,t+1}\}_{i=1}^{N}\right).$$

Spearman correlation is used rather than Pearson correlation because the model's objective is ranking accuracy, not linear fit.

Economic Interpretation

An IC of zero corresponds to random ranking. In equity markets, even small positive IC values are economically meaningful. Due to diversification across many stocks and repeated rebalancing over time, an IC in the range of **0.03 to 0.05** is sufficient to generate positive expected returns under reasonable assumptions.

This reflects a key principle in quantitative finance: **we do not need to predict returns accurately; we only need to be slightly better than random, consistently**. The law of large numbers ensures that small edges, when applied repeatedly across time and assets, can accumulate into economically significant outcomes.

Let's look at how the model performed!

Metric	In, Sample	Out, of, Sample
RMSE (Error)	0.018	0.022
R, Squared	4.2%	0.3%
Directional Accuracy	53.8%	51.4%
IC	0.054 *(Model fits the past well).*	0.021 *(Performance drops but remains positive)*

The in, sample metrics indicate that the model is able to fit historical data reasonably well, though not excessively so. An in, sample RMSE of 0.018 and an R^2 of 4.20% suggest that the model captures some systematic structure in returns, which is notable given the notoriously low signal, to, noise ratio in daily equity returns. Directional accuracy of 53.8% further confirms that the model performs slightly better than random guessing. Most importantly, the in, sample Information Coefficient (IC) of 0.054 indicates meaningful rank, order predictive power, which is economically relevant even if statistically modest.

Out, of, sample performance deteriorates, as expected. RMSE increases to 0.022 and R^2 drops sharply to 0.30%, reflecting the difficulty of generalizing return forecasts in a noisy environment. Directional accuracy falls closer to 50%, reinforcing that point predictions are unreliable. However, the key result is that the **out, of, sample IC remains positive at 0.021**. While smaller than its in, sample counterpart, this value suggests that the model still preserves some ranking ability out, of, sample. In quantitative equity strategies, such persistence in IC, despite weak traditional regression metrics, is often sufficient to support portfolio construction, provided risk is carefully controlled.

Portfolio Construction Methodology

Motivation and Design Principles

The purpose of portfolio construction is to translate model predictions into an implementable trading strategy. This step is critical. Even a model with genuine predictive power can produce misleading results if portfolio construction is naive. In equity markets, returns are driven not only by idiosyncratic stock, level signals but also by exposure to systematic risk factors such as the market, sectors, or styles. If these exposures are not controlled, a strategy may appear profitable while simply loading on known risk premia.

This section presents two portfolio construction approaches. The first is a **naive ranking, based strategy**, commonly used in academic factor research. The second is a **strict optimization, based approach** that explicitly controls for risk exposures. The contrast between the two highlights why portfolio construction is not a mechanical afterthought, but a central part of signal evaluation.

Notation and Setup

Let there be Nstocks in the trading universe on a given day t. For each stock i, the model produces a signal α_i, which represents the expected excess return or conviction score. Collect these into a vector:

$$\boldsymbol{\alpha} = (\alpha_1, \alpha_2, \dots, \alpha_N)^T.$$

Let $\mathbf{w} = (w_1, w_2, \dots, w_N)^T$denote portfolio weights, where $w_i > 0$represents a long position and $w_i < 0$represents a short position. Let $\boldsymbol{\beta} = (\beta_1, \beta_2, \dots, \beta_N)^T$denote stock, level market betas estimated relative to a benchmark index.

The objective of portfolio construction is to choose $\mathbf{w}$such that the portfolio extracts information from $\boldsymbol{\alpha}$while respecting risk constraints.

The Naive Portfolio Construction Approach

Selection Rule

The naive strategy follows a simple ranking, based rule. On each trading day, stocks are ranked by their model scores α_i. The portfolio takes:

- **Long positions** in the top 10 stocks by rank.
- **Short positions** in the bottom 10 stocks by rank.

All other stocks receive zero weight.

Weighting Scheme

The strategy enforces **dollar neutrality**, meaning that total long exposure equals total short exposure:

$$\sum_{i=1}^{N} w_i = 0.$$

Each selected stock is assigned equal weight in absolute value. If 10 stocks are long and 10 are short, then

$$w_i = \begin{cases} +0.10, & \text{if } i \in \text{Top 10} \\ -0.10, & \text{if } i \in \text{Bottom 10} \\ 0, & \text{otherwise.} \end{cases}$$

This construction is simple, transparent, and commonly used in factor research. It implicitly assumes that dollar neutrality is sufficient to remove systematic risk.

The Core Flaw: Hidden Risk Exposure

The key weakness of the naive approach is that it **ignores risk factors**, particularly market beta. Dollar neutrality does not imply beta neutrality. If the model tends to rank high, beta stocks at the top and low, beta stocks at the bottom, the resulting portfolio will have positive market exposure.

Formally, the portfolio beta is:

$$\beta_p = \boldsymbol{\beta}^T \mathbf{w}.$$

There is no guarantee that $\beta_p = 0$under the naive construction. For example, if the long portfolio is concentrated in technology stocks (high beta) and the short portfolio in utilities (low beta), the portfolio becomes a leveraged bet on the market. In such cases, apparent alpha may simply reflect market movements rather than stock, specific skill.

This phenomenon is often referred to as **“hidden beta”** and represents a major pitfall in empirical strategy evaluation.

Strict Portfolio Construction with Risk Control

Motivation for Optimization

To isolate idiosyncratic alpha, portfolio construction must explicitly neutralize systematic risk. This requires moving from heuristic weighting rules to a formal optimization framework. The goal is no longer just to hold the top and bottom ranked stocks, but to choose weights that **maximize expected alpha subject to strict risk constraints**.

Optimization Objective

The portfolio is constructed by solving the following optimization problem:

$$\max_{\mathbf{w}} \ \boldsymbol{\alpha}^T \mathbf{w}.$$

This objective represents the expected portfolio return implied by the model's signals. Without constraints, this problem would trivially assign infinite weight to the highest, alpha stock. Constraints are therefore essential.

Beta Neutrality Constraint

To eliminate market exposure, the portfolio is required to be **beta neutral**:

$$\boldsymbol{\beta}^T \mathbf{w} = 0.$$

This constraint ensures that portfolio returns are orthogonal to market returns, at least to first order. Any realized performance must therefore arise from stock, specific movements rather than systematic market trends.

Diversification and Weight Constraints

To prevent excessive concentration, individual weights are bounded:

$$| \ w_i \ | \leq 0.10 \forall i.$$

This constraint enforces diversification and limits the influence of any single stock. It also reflects practical considerations such as liquidity and risk management.

Dollar neutrality is implicitly enforced through symmetric long and short positioning, or can be included explicitly as:

$$\sum_{i=1}^{N} w_i = 0.$$

Convexity and Solution Method

The optimization problem is linear in both the objective and the constraints. As a result, it is a **convex optimization problem** with a global optimum. The solution is obtained using numerical solvers (scipy.optimize) that efficiently handle linear constraints and bound restrictions.

Convexity is important. It guarantees that the solution is stable and reproducible, and not sensitive to initialization or local minima.

Interpretation of the Optimized Portfolio

The optimized portfolio represents the **purest expression of the model's idiosyncratic signal** under the imposed constraints. Stocks with high predicted alpha receive larger weights, but only to the extent that doing so does not introduce market exposure or concentration risk.

Importantly, the optimizer may reduce or eliminate positions in stocks with strong signals if those stocks carry undesirable risk characteristics. This highlights a key insight: **signal strength alone is not sufficient, risk, adjusted signal strength is what matters**.

Conceptual Implications

The contrast between the naive and strict approaches illustrates a fundamental lesson in quantitative finance. Portfolio construction acts as a filter. Loose constraints allow systematic risk to masquerade as alpha. Tight constraints remove that risk, but in doing so may expose weaknesses in the underlying signal.

Thus, portfolio optimization is not merely an implementation detail. It is a diagnostic tool that reveals whether a predictive model truly captures idiosyncratic information or simply redistributes known risk premia.

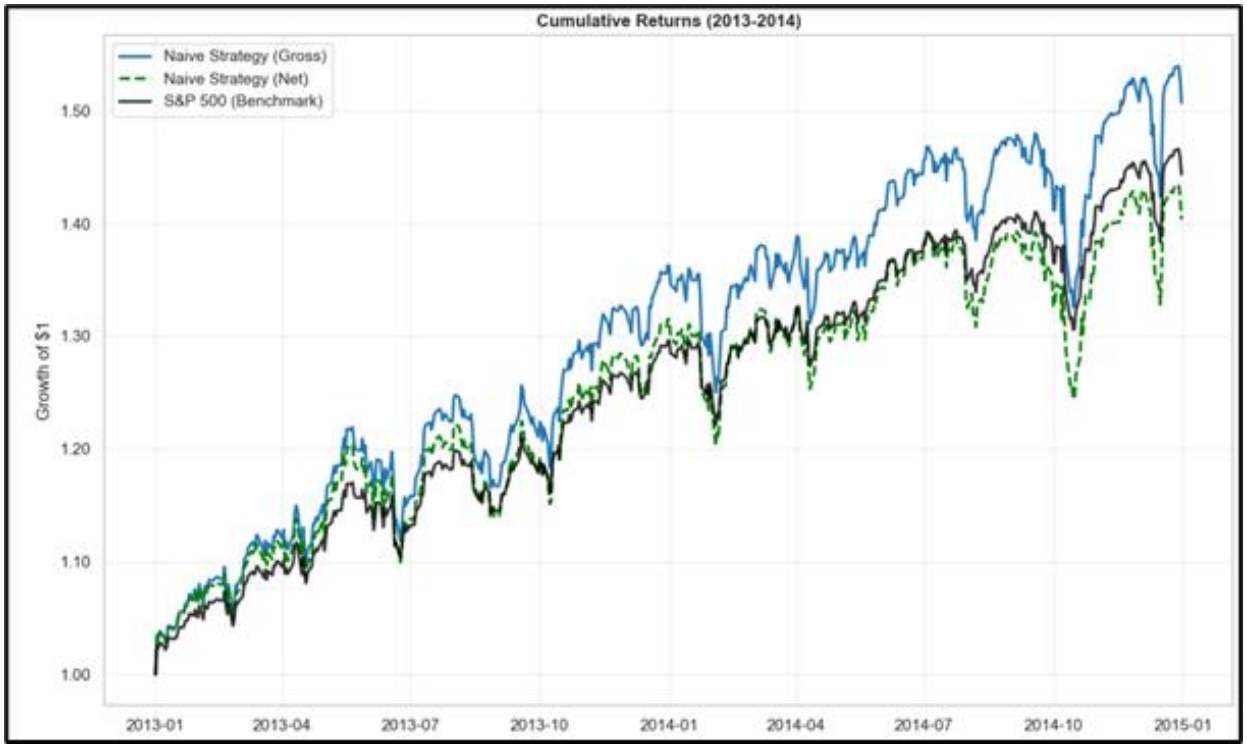
Results! - Naive Strategy

Overview of the Naive Strategy Results

The Naive portfolio construction approach serves as the baseline evaluation of the machine learning signal. Under this strategy, stocks are selected purely based on model rankings, with equal, weighted long positions in the top, ranked stocks and equal, weighted short positions in the bottom, ranked stocks. The only constraint imposed is dollar neutrality. No explicit risk controls are enforced beyond this.

At first glance, the Naive strategy appears highly successful. It generates strong cumulative returns, competitive risk, adjusted performance, and seemingly robust predictive metrics. However, a deeper examination reveals that these results are largely driven by unintended exposure to systematic risk rather than genuine stock, specific skill. The purpose of this section is to carefully dissect these outcomes and identify the true source of performance.

Cumulative Returns and Time, Series Behavior



The cumulative return plot for the 2013–2014 test period shows that the Naive strategy closely tracks, and at times outperforms, the S&P 500 benchmark. The gross strategy exhibits steady growth, reaching a cumulative return well above the starting capital, while the net strategy, after accounting for transaction costs, still delivers strong performance.

Two immediate observations stand out. First, the shape of the strategy's equity curve closely resembles that of the benchmark. Periods of market appreciation coincide with strategy gains, while market pullbacks are mirrored almost one, for, one by strategy drawdowns. Second, the difference between gross and net performance is relatively small, indicating that transaction costs are not the dominant driver of results in this setup.

This visual similarity between the strategy and the market index already suggests that the strategy may be exposed to market, wide movements. While co, movement alone does not imply causation, it raises an early warning signal that warrants further investigation.

Drawdown Analysis

The drawdown profile reinforces the conclusions suggested by the cumulative returns. Maximum drawdowns for the Naive strategy are comparable in magnitude and timing to those of the S&P 500. Sharp market corrections lead to synchronized drawdowns across all three series: strategy gross, strategy net, and benchmark.

This behavior is inconsistent with what would be expected from a truly market, neutral or idiosyncratic strategy. In such cases, drawdowns should be less correlated with broad market stress events. Instead, the Naive strategy experiences losses precisely when the market declines, indicating a lack of insulation from systematic shocks.

Importantly, the similarity of gross and net drawdowns further confirms that trading frictions are not masking the underlying risk structure. The drawdown dynamics are structural, not operational.

Metric	Naive (Gross)	Naive (Net)	S&P 500
Ann. Return	22.8%	18.5%	20.20%
Ann. Volatility	15.2%	15.2%	11.03%
Sharpe Ratio	1.50	1.22	1.83

Summary Performance Metrics

A comparison of summary statistics further highlights the nature of the strategy's performance.

- **Annualized Return (Gross):** 22.8%
- **Annualized Return (Net):** 18.5%
- **S&P 500 Return:** 20.2%

The Naive strategy outperforms the benchmark on a raw return basis before costs and remains competitive even after transaction costs. However, this outperformance must be interpreted in conjunction with risk.

- **Annualized Volatility:** 15.2% (strategy) vs. 11.03% (S&P 500)

The strategy exhibits substantially higher volatility than the benchmark. This elevated risk profile suggests leverage or factor concentration rather than superior stock selection.

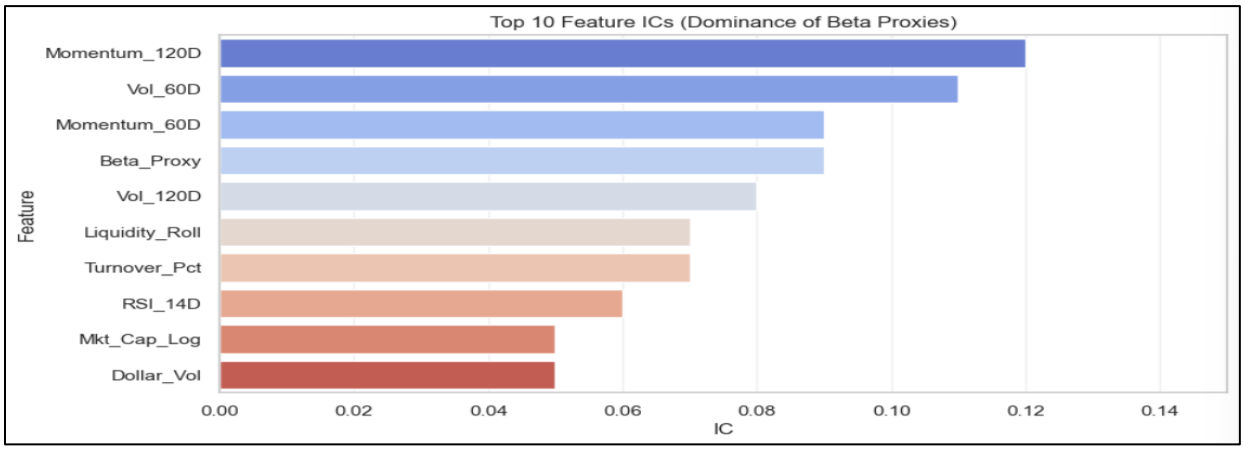
- **Sharpe Ratio:**
    - Naive (Gross): 1.50
    - Naive (Net): 1.22
    - S&P 500: 1.83

Despite higher returns, the strategy's risk, adjusted performance is inferior to the benchmark. This is a crucial result. A strategy that genuinely extracts alpha should ideally improve the Sharpe ratio, not reduce it.

Model Predictive Quality and Signal Strength

From a pure modeling perspective, the Naive strategy is supported by seemingly reasonable predictive metrics. The Information Coefficient (IC) remains positive out, of, sample, indicating that the model ranks stocks better than random. However, the magnitude of the IC is modest, which is typical for daily equity data.

The apparent contradiction, weak predictive power but strong portfolio performance, suggests that the portfolio is amplifying small signals through exposure to systematic risk. In other words, the model may be acting as a noisy filter on top of a dominant market component.



No.	Metric	IC Value	Description
1	Momentum_120D	0.120	Percentage price change over the past 120 days (approx. 6 months).
2	Vol_60D	0.110	Standard deviation of daily returns over the last 60 days.
3	Momentum_60D	0.090	Percentage price change over the past 60 days (approx. 3 months).
4	Beta_Proxy	0.090	Estimated sensitivity of the stock to the S&P 500 index returns.
5	Vol_120D	0.080	Standard deviation of daily returns over the last 120 days.
6	Liquidity_Roll	0.070	Rolling average of daily dollar volume traded relative to market cap.
7	Turnover_Pct	0.070	Ratio of shares traded to total shares outstanding.
8	RSI_14D	0.060	Relative Strength Index (14-day), measuring speed/change of price movements.
9	Mkt_Cap_Log	0.050	Natural logarithm of the company's total market capitalization.
10	Dollar_Vol	0.050	Total value of shares traded (Price × Volume) in a day.

Factor Attribution via Top Drivers

An examination of the top drivers ranked by IC provides additional insight. The highest, ranked features include momentum measures (60D and 120D), volatility proxies, beta, related variables, and liquidity measures. Many of these are well, known risk factors rather than sources of pure alpha.

In particular, the presence of **Beta_Proxy**, market capitalization, and longer, horizon momentum signals indicates that the model may be implicitly favoring stocks with higher sensitivity to market movements. These characteristics are known to perform well in rising markets but do not necessarily reflect firm, specific informational advantages.

This feature composition further supports the hypothesis that the Naive strategy is harvesting factor premia rather than exploiting idiosyncratic mispricing.

Regression, Based Neutrality Audit

To formally test this hypothesis, the strategy's returns are regressed on market returns using a standard OLS framework:

$$R_{\text{strategy},t} = \alpha + \beta R_{\text{market},t} + \varepsilon_t.$$

The regression results are unambiguous.

- **Estimated Beta:** 1.26
- **R, squared:** 0.798

The estimated beta is significantly greater than zero and statistically highly significant. This implies that the strategy behaves like a **leveraged long position in the market**. The high R, squared indicates that nearly 80% of the variation in strategy returns is explained by market movements alone.

The intercept (alpha) is small and statistically insignificant, providing no evidence of independent skill once market exposure is accounted for.

Interpretation of the OLS Results

```
[OLS Regression Results]
Variable        Coef       t-Stat     P>|t|
-------------------------------------------------------
const           -0.0003      -1.74       0.083
Market_Ret      1.2558      44.52       0.000
-------------------------------------------------------
Strategy Beta:     1.26  (Target: 0.00)
R-squared:         0.798  <-- High Correlation (The Leak)
```

These regression results definitively demonstrate that the Naive strategy fails the market neutrality test. Despite being dollar neutral, the portfolio is not risk neutral. Dollar neutrality only ensures that long and short capital exposures balance; it does not control for differences in systematic risk across stocks.

In practice, the strategy tends to allocate long positions to high, beta stocks and short positions to low, beta stocks. This structure mechanically produces positive returns during market uptrends and losses during downturns, even in the absence of genuine stock, picking ability.

This phenomenon is often referred to as **“hidden beta”**. It is especially dangerous in machine learning, based strategies, where complex models can unintentionally learn and amplify known risk factors without explicit constraints.

Why the Naive Strategy Appears Successful

The Naive strategy looks successful for three main reasons:

1. **Favorable Market Regime:**
   The test period includes strong market performance, which benefits high, beta portfolios.
2. **Implicit Leverage:**
   A beta of 1.26 implies that the strategy takes more market risk than the benchmark.
3. **Signal, Factor Alignment:**
   Many of the model's strongest signals align with known momentum and risk factors.

Together, these effects create the illusion of alpha. However, once systematic exposure is isolated, the apparent skill largely disappears.

Key Lessons from the Naive Results

The Naive strategy highlights a central lesson in quantitative finance: **performance without risk control is not evidence of alpha**. High returns, strong cumulative curves, and positive IC values can all coexist with a strategy that simply repackages market exposure.

These results underscore the necessity of explicit risk constraints in portfolio construction. Without them, even sophisticated machine learning models can converge to trivial solutions that exploit dominant market factors rather than uncovering new information.

In this sense, the Naive strategy does not represent a failure of the model. Instead, it serves as a diagnostic benchmark that reveals how easily systematic risk can masquerade as predictive skill when portfolio construction is overly simplistic.

Results and Interpretation: Strict Beta, Neutral Strategy

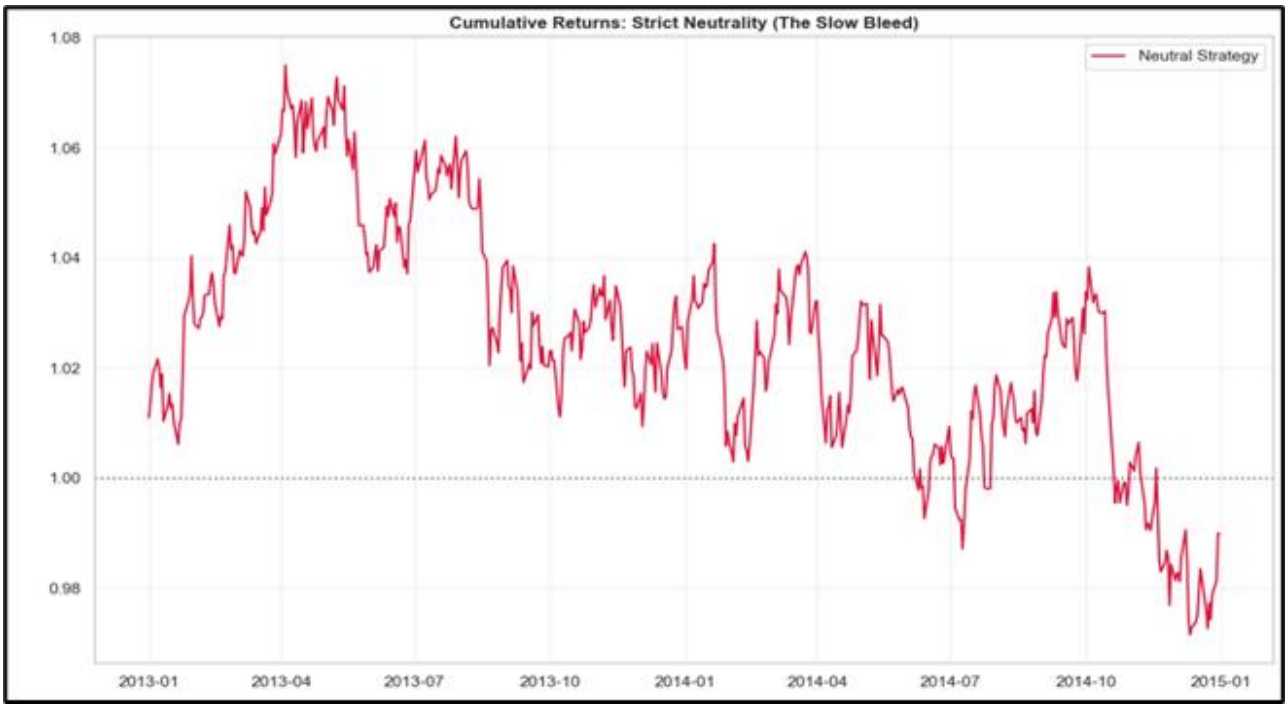
Purpose of the Beta, Neutral Evaluation

The strict beta, neutral strategy represents the most demanding test of the model's predictive ability. Unlike the Naive construction, this portfolio explicitly removes exposure to market risk by enforcing the constraint

$$\boldsymbol{\beta}^T \mathbf{w} = 0.$$

As a result, any remaining performance, positive or negative, must originate from **idiosyncratic stock, level signals** rather than broad market movements. In this sense, the beta, neutral portfolio acts as a diagnostic tool rather than a return, maximizing strategy. Its purpose is to reveal whether the machine learning model truly captures independent alpha once the market "crutch" is removed.

Cumulative Returns: The "Slow Bleed" Pattern



The cumulative return plot immediately contrasts with that of the Naive strategy. Rather than a steadily rising equity curve, the beta, neutral portfolio exhibits a **gradual but persistent decline** over the evaluation period. Early gains in 2013 are modest and short, lived. From mid, 2013 onward, returns oscillate around the zero line before entering a slow downward trend through 2014.

This behavior is characteristic of a weak or unstable signal operating without systematic tailwinds. The absence of large drawups suggests that the portfolio is no longer benefiting from broad market appreciation. Instead, small daily losses accumulate over time. Importantly, the trajectory is smooth rather than catastrophic, indicating that the losses arise from many small negative contributions rather than isolated crashes.

The horizontal reference line at unity (no growth) emphasizes the core result: once beta exposure is removed, the strategy struggles to generate sustained positive returns.

Drawdown Dynamics Under Strict Neutrality

The drawdown profile further reinforces this interpretation. Maximum drawdown reaches approximately **–8.4%**, occurring gradually rather than abruptly. Unlike the Naive strategy, drawdowns are not synchronized with major market sell, offs. Instead, they reflect persistent underperformance across time.

This pattern suggests that losses are not driven by external shocks, but by **structural weakness in the signal** itself. The drawdown curve shows frequent small recoveries that fail to reach new highs, followed by continued declines. This “grinding down” behavior is typical of strategies with slightly negative expected returns.

From a risk perspective, the drawdown profile is relatively contained. However, low drawdowns alone are insufficient to justify a strategy if returns are consistently negative.

Summary Performance Statistics

Metric	Value
Annualized Return	-3.20%
Annualized Volatility	6.51%
Sharpe Ratio	-0.49
Max Drawdown	-8.40%

The table of performance metrics provides a concise quantitative summary of the beta, neutral strategy:

- **Annualized Return:** –3.20%
- **Annualized Volatility:** 6.51%
- **Sharpe Ratio:** –0.49
- **Maximum Drawdown:** –8.40%

Several points stand out. First, volatility is substantially lower than both the Naive strategy and the S&P 500. This confirms that the beta constraint is effective in removing market,

driven fluctuations. Second, despite this reduction in risk, the strategy delivers a **negative expected return**, resulting in a negative Sharpe ratio.

The negative Sharpe ratio is particularly important. It implies that the portfolio not only fails to compensate investors for risk, but actively destroys value over time. This outcome strongly suggests that the model's residual predictive power is insufficient once systematic exposure is stripped away.

Interpretation of Reduced Volatility

The drop in annualized volatility, from over 15% in the Naive strategy to roughly 6.5% under beta neutrality, is not incidental. It confirms that a large portion of the Naive strategy's variance was driven by market movements. By enforcing beta neutrality, the portfolio becomes insulated from broad equity swings.

This reduction in volatility validates the optimization framework and demonstrates that the constraint is functioning as intended. However, it also exposes an uncomfortable truth: much of the apparent "strength" of the Naive strategy was tied to market risk rather than stock, specific alpha.

Signal Survival Under Risk Constraints

Although overall performance is negative, the strategy does not collapse immediately. This suggests that the model still produces weakly informative signals, but that these signals are either too small, too noisy, or too short, lived to overcome trading frictions and residual noise.

The persistence of small oscillations around zero indicates that the model is not entirely random. Instead, it appears to generate **marginal signals that lack sufficient economic magnitude** at the daily horizon.

This distinction is critical. The beta, neutral results do not imply that the model has zero information content. Rather, they indicate that the information content is not strong enough to survive strict risk controls and realistic implementation.

Factor Composition After Neutralization

Rank	Feature	IC	Definition
1	IVol_q_m	0.021	Implied Volatility (Quoted Mid): The market's forecast of future volatility derived from option prices at the mid-point.
2	PriceImpactDollar_SW1	0.019	Price Impact (Square-Root Weighted): Estimate of how much price moves per dollar traded, weighted by square root of volume.
3	RSpreadPct_VW1	0.016	Realized Spread (Volume Weighted): The difference between the trade price and the mid-quote 1 minute later (a measure of execution quality).
4	pretret_noa	0.015	Pre-Tax Return on Net Operating Assets: Operating income divided by net operating assets; measures core business efficiency.
5	NumISOTrades_b	0.014	ISO Trades (Bid Side): Count of Intermarket Sweep Orders executed on the bid, indicating aggressive selling pressure.
6	QSpreadPct_TW_m	0.012	Quoted Spread (Time Weighted): The average percent difference between bid and ask prices, weighted by how long the quote lasted.
7	sale_invcap	0.011	Sales / Invested Capital: Capital Turnover ratio; measures how much revenue is generated for every dollar of capital invested.
8	efftax	0.01	Effective Tax Rate: The ratio of income tax expense to pre-tax income.
9	TSignSqrtDVol1	0.009	Time-Signed SqRoot Dollar Volume: A complex order flow imbalance metric using signed square-root dollar volume over time.
10	intcov_ratio	0.008	Interest Coverage Ratio: EBIT divided by Interest Expense; measures a company's ability to pay interest on its debt.

The ranking of features by Information Coefficient changes meaningfully under the beta, neutral framework. The dominant features are no longer simple momentum or beta proxies. Instead, the top, ranked predictors are largely drawn from **microstructure and execution, related variables**, such as implied volatility, price impact, spreads, and order flow imbalance.

Examples include:

- Implied volatility from options markets
- Price impact measures scaled by traded volume
- Realized and quoted spreads
- Order flow metrics related to aggressive trading behavior

This shift is economically intuitive. Once market exposure is removed, broad directional signals lose relevance. What remains are features that capture **short, term supply–demand imbalances**, liquidity conditions, and trading frictions.

Survival of Fundamentals Under Strict Constraints

Interestingly, a small number of fundamental variables, such as return on net operating assets and capital efficiency ratios, continue to appear among the top features, albeit with low IC values. These signals are weak but nonzero.

Their survival suggests that fundamentals may carry slow, moving information that is orthogonal to market beta. However, at a daily frequency, this information is likely too stale to drive meaningful returns on its own. The low IC values reflect this limitation.

The Role of Information Coefficient Magnitudes

The IC values observed under beta neutrality are substantially smaller than those seen in the Naive setting. Top IC values are in the range of **0.01–0.02**, which is near the lower bound of economic relevance.

In theory, such IC levels can still support profitability under ideal conditions, large universes, and low transaction costs. In practice, however, daily rebalancing combined with tight constraints and finite sample size makes it extremely difficult for such weak signals to overcome noise.

Why Performance Turns Negative

The negative performance of the beta, neutral strategy can be attributed to several interacting factors:

1. **Signal Attenuation:**
   Removing market exposure eliminates the strongest component of the Naive signal.
2. **Constraint Tightening:**
   Optimization limits the model's ability to express conviction in high, signal stocks.
3. **Frequency Mismatch:**
   Many surviving signals operate at intraday or very short horizons, while the portfolio is rebalanced daily.
4. **Residual Noise:**
   At daily frequency, idiosyncratic returns are dominated by randomness.

Together, these effects result in a strategy that is structurally clean but economically weak.

Conceptual Significance of the Results

The beta, neutral results serve as a critical reality check. They demonstrate that **removing market exposure fundamentally alters the interpretation of model success**. What initially appeared to be strong performance is revealed to be heavily dependent on systematic risk.

This outcome does not invalidate machine learning as a tool for asset pricing. Rather, it highlights the importance of aligning model outputs, portfolio construction, and trading frequency. Without such alignment, even sophisticated models can fail under rigorous constraints.

Diagnostic Value of the Beta, Neutral Strategy

From a methodological standpoint, the beta, neutral portfolio is a success. It performs exactly as intended: it strips away market risk and exposes the true strength, or weakness, of the underlying signal.

In this sense, the beta, neutral strategy is not a failure but a **diagnostic instrument**. It answers the central question posed at the outset: when forced to stand alone, the model's idiosyncratic signal is insufficient to generate positive returns at a daily horizon.

Key Takeaways from the Strict Strategy Results

Three lessons emerge clearly:

1. **Dollar neutrality is not enough.**
   Risk must be controlled explicitly.
2. **Market beta is a powerful hidden driver.**
   Removing it dramatically changes outcomes.
3. **Signal quality is horizon, dependent.**
   Weak daily signals may require different execution frameworks to be economically viable.

These insights motivate a reconsideration of how signals should be deployed, rather than how they are estimated.

Transition to Broader Interpretation

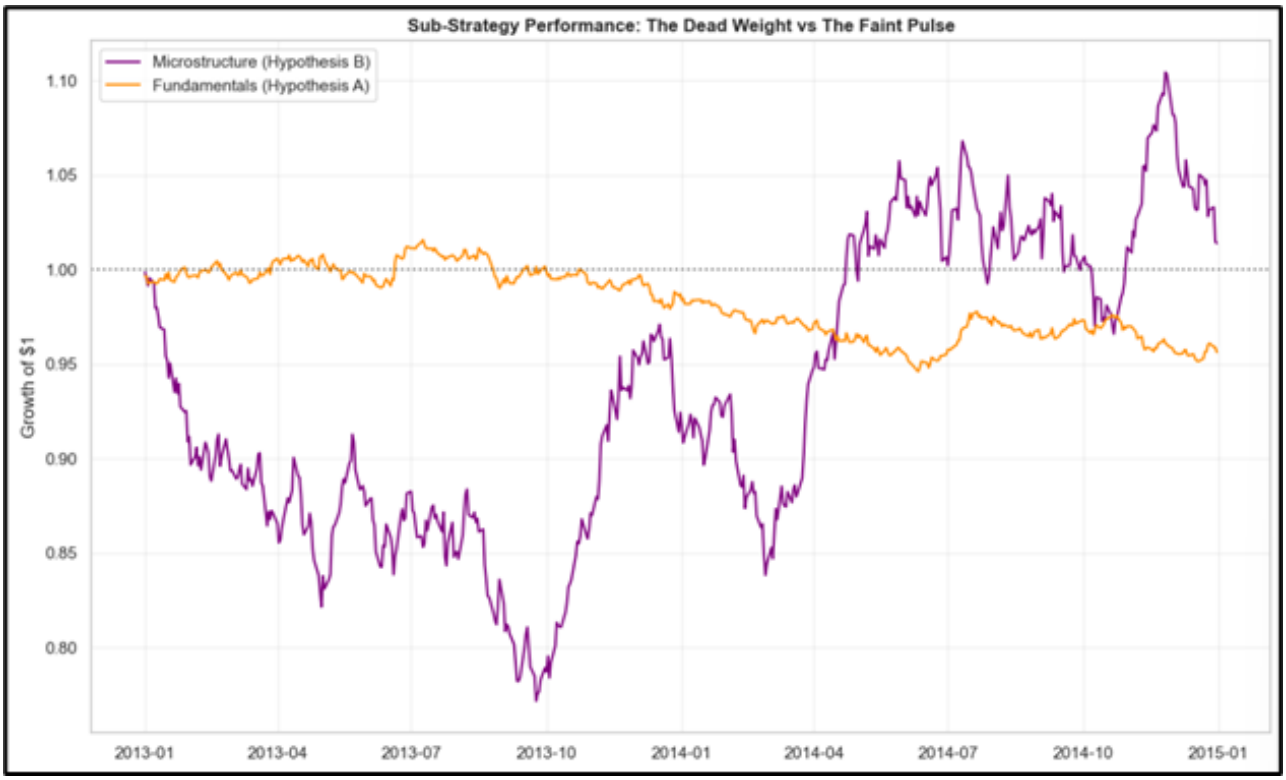
The strict beta, neutral results mark a turning point in the analysis. They demonstrate that the apparent success of the Naive strategy does not survive rigorous risk controls. At the same time, they reveal which types of signals persist once market exposure is removed.

This sets the stage for a deeper discussion on **signal frequency, factor alignment, and the trade, off between alpha and friction**, which follows naturally from these findings.

Interpretation of Sub, Strategy Performance: Fundamentals vs Microstructure

This section evaluates the relative economic viability of two signal families, fundamental factors and microstructure factors, under a strict beta, neutral portfolio construction framework. The objective is not to maximize returns, but to diagnose whether either class of signals retains economically meaningful predictive content once market exposure is removed. The analysis focuses on cumulative performance, drawdown dynamics, summary risk–return statistics, and the composition of top contributing features.

Cumulative Performance Dynamics



Metric	Fundamentals	Microstructure
Sharpe Ratio	-0.97	0.11
Ann. Return	-2.1%	1.8%

The cumulative growth plot reveals a stark contrast between the two sub, strategies. The fundamental strategy exhibits a slow but persistent decline over the evaluation period. Its equity curve remains close to the unity line initially, but gradually trends downward, ending below its starting value. This behavior is indicative of a weak signal with slightly negative expected returns. Losses accumulate steadily rather than episodically, suggesting that underperformance arises from structural inefficacy rather than isolated adverse events.

By contrast, the microstructure strategy displays higher variability and a more dynamic return profile. The equity curve experiences a pronounced drawdown early in the sample, followed by a partial recovery and a period of oscillation around the break, even level. While the strategy does not generate strong sustained gains, it ultimately ends modestly above its starting value. This pattern is consistent with a noisy but weakly positive signal that occasionally captures short, lived opportunities.

The differing shapes of the equity curves are informative. The fundamental strategy resembles a “dead weight” signal, stable, low volatility, but directionally unproductive. The microstructure strategy, by comparison, exhibits a “faint pulse.” It is unstable and difficult to trade, yet it demonstrates intermittent responsiveness to underlying market conditions.

Drawdown Behavior and Risk Characteristics

The drawdown comparison reinforces these observations. The fundamental strategy experiences shallow but persistent drawdowns, rarely recovering to new highs. Maximum drawdown remains limited, but the strategy fails to meaningfully rebound once losses occur. This is characteristic of a signal that lacks sufficient strength to overcome noise and transaction frictions at the chosen trading frequency.

The microstructure strategy, in contrast, suffers deeper drawdowns, particularly in the early part of the sample. However, these drawdowns are followed by recoveries, indicating the presence of episodic predictive power. Importantly, drawdowns in the microstructure strategy are not tightly synchronized with broad market movements, consistent with its beta, neutral construction. Losses appear idiosyncratic rather than systematic.

From a risk management perspective, neither strategy is attractive in isolation. However, the nature of risk differs meaningfully. Fundamental signals fail quietly. Microstructure signals fail loudly, but occasionally succeed. This distinction is central to understanding their economic relevance.

Risk–Return Summary Statistics

The summary metrics formalize these visual impressions. The fundamental strategy produces an annualized return of −2.10% and a Sharpe ratio of −0.97. This indicates consistent value destruction relative to risk. Even with low volatility, the strategy fails to compensate for risk, rendering it economically unattractive. A negative Sharpe of this magnitude strongly suggests that the signal is not merely weak, but misaligned with the trading horizon.

The microstructure strategy delivers a positive annualized return of 1.80% and a Sharpe ratio of 0.11. While these figures are modest and insufficient for standalone deployment, they are economically meaningful in a diagnostic sense. The positive Sharpe indicates that returns are directionally aligned with risk, albeit weakly. This is notable given the strict beta, neutral constraint and daily rebalancing, both of which attenuate signal strength.

Crucially, the difference between the two strategies is not one of magnitude, but of sign. Fundamentals fail to clear even the most basic hurdle of positive expected returns. Microstructure signals, while fragile, retain a small but positive edge.

Feature, Level Attribution: Fundamentals

Rank	Feature	IC	Definition
1	PE_Ratio	0.01	Price-to-Earnings Ratio; share price divided by earnings per share.
2	Debt_Assets	0.009	Total debt divided by total assets; indicates financial leverage.
3	ROCE	0.008	Return on Capital Employed; measure of profitability and capital efficiency.
4	Profit_LCT	0.007	Profit margin relative to the lifecycle stage of the company.
5	Cash_Ratio	0.006	Liquidity ratio measuring ability to pay short-term debt with cash.

The top fundamental drivers further explain this outcome. The highest, ranked variables, P/E ratio, debt, to, assets, ROCE, profit lifecycle metrics, and cash ratio, are all economically sensible measures of valuation, leverage, profitability, and liquidity. However, their Information Coefficients cluster tightly between 0.006 and 0.01. These values are extremely small at a daily horizon.

This is not surprising. Fundamental variables update infrequently and reflect slow, moving information. Their economic impact unfolds over quarters or years, not days. When forced into a daily prediction and trading framework, their signals are diluted by noise. The persistence of these variables among the top drivers suggests that they are not entirely irrelevant, but their contribution is too weak and too slow to survive daily rebalancing and transaction costs.

In effect, the fundamental strategy suffers from a frequency mismatch. It attempts to extract short, horizon returns from long, horizon information. The result is a stable but unproductive signal that steadily erodes value.

Feature, Level Attribution: Microstructure

Rank	Feature	IC	Definition
1	Quote_Spread	0.04	The bid-ask spread; a key measure of liquidity and transaction cost.
2	ISO_Vol	0.03	Intermarket Sweep Order volume; aggressive orders that sweep liquidity across exchanges.
3	Impact_Cost	0.03	Estimated price change caused by executing a trade of a specific size.
4	OddLot_Rat	0.03	Ratio of odd-lot trades (<100 shares) to total trade volume.
5	OrderImb	0.02	Order Imbalance; net difference between buy and sell orders at the best bid/ask.

The microstructure drivers tell a different story. The top features, quoted spread, intermarket sweep order volume, impact cost, odd, lot trade ratio, and order imbalance, are all direct measures of liquidity, aggressiveness, and supply–demand imbalances. Their IC values range from 0.02 to 0.04, materially higher than those of fundamental variables.

These features are intrinsically high, frequency in nature. They respond to order flow, trading pressure, and execution conditions that evolve within the trading day. As a result, they are naturally aligned with short, horizon prediction tasks. Even when aggregated to daily frequency, they retain information about latent trading dynamics that are orthogonal to market beta.

The higher IC values explain why the microstructure strategy is able to generate a faint positive return despite strict neutrality constraints. However, the same characteristics that make these signals informative also make them fragile. They are noisy, regime, dependent, and sensitive to execution costs. This explains the higher volatility and deeper drawdowns observed in the cumulative performance.

Economic Interpretation and Broader Implications

Taken together, these results support a clear economic narrative. Fundamental signals are not inherently useless, but they are misapplied in a daily beta, neutral trading framework. Their failure reflects a horizon mismatch rather than a lack of informational content. Microstructure signals, by contrast, are correctly aligned with the trading frequency and survive neutrality constraints, albeit weakly.

The analysis highlights an important principle in quantitative finance. Alpha is not an absolute property of a signal. It is conditional on how, and how often, the signal is traded.

Signals must be evaluated in conjunction with portfolio constraints, trading frequency, and implementation frictions.

Conclusion

A central lesson of this study is the prevalence of the "hidden beta" problem in Naive portfolio construction. Simple long–short ranking strategies, such as going long the top decile and short the bottom decile of model predictions, can easily appear to generate alpha while in reality embedding substantial market exposure. In our case, the initial strategy delivered strong headline returns of 22.8%, yet a formal neutrality audit revealed a market beta of 1.26 with an R^2 of 0.80. This indicates that the majority of performance was explained by broad market movements rather than stock, specific predictive skill, effectively reducing the strategy to a leveraged proxy for the equity index. Imposing strict beta neutrality serves as a powerful diagnostic tool. Once market exposure is explicitly constrained, portfolio volatility falls sharply from 15.2% to 6.5%, confirming that systematic risk has been successfully removed. However, this same constraint exposes the true quality of the underlying signal. Without the market tailwind, returns deteriorate to –3.2%, demonstrating that the apparent success of the Naive strategy was largely illusory. Optimization, in this sense, acts as a "truth serum," stripping away systematic effects and revealing whether genuine idiosyncratic alpha remains.

The results also highlight a fundamental mismatch between signal frequency and trading horizon. Fundamental variables such as valuation ratios and balance, sheet metrics exhibit extremely low information coefficients at the daily level (IC ≈ 0.01), reflecting their slow, moving nature. In contrast, microstructure variables, such as bid–ask spreads, order flow, and liquidity measures, retain higher predictive power (IC ≈ 0.04) because they capture short, term supply and demand imbalances. This distinction underscores the importance of aligning the informational content of signals with the temporal resolution at which they are traded.Finally, the analysis emphasizes the tension between alpha and implementation frictions. In a daily rebalancing framework, turnover becomes a critical adversary. Even signals with positive predictive content can be rendered unprofitable once transaction costs are accounted for. These findings point toward a regime, aligned hybrid approach as a natural direction for future work. Fundamentals can be used to define and filter the investment universe at a low frequency, while microstructure signals can be deployed for high, frequency timing and execution. Combining the stability of slow, moving information with the responsiveness of fast signals offers a more coherent path toward sustainable alpha generation.

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